
CS771 Minor Assignment 3

Group: Deepminds
Melbo's 'Mazing Mixtures

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Abstract

This report presents our analysis of the Mixed Regression approach using the EM algorithm for predicting ground-truth ozone concentrations (Ref . 03(ppb)) from low-cost sensor data. We study the effect of the number of EM iterations and mixture components on prediction accuracy, perform a qualitative analysis of component specialisation, conduct a feature ablation study, and compare the mixture regression approach against `sklearn.tree.DecisionTreeRegressor` baselines. All experiments use the `DD1(Oct).csv` dataset with the code template provided in `lec16.ipynb`.

Experimental Setup

Dataset and Preprocessing

We use the `DD1(Oct).csv` air-quality dataset. Following the assignment instructions, we:

1. Filter to *valid* rows (`Valid == 1`).
2. Select columns: Time (column 0), Ref . 03(ppb) (column 1), Temp(C) (column 3), RH(%) (column 4), `o3op1(mV)` (column 7), `o3op2(mV)` (column 8).
3. Convert Time to the hour-of-day integer.
4. Split 80/20 chronologically (no random split for time-series data).
5. Standardise features using training-set mean and standard deviation.

The label is Ref . 03(ppb) (column index 1). The five input features are: Time, Temp(C), RH(%), `o3op1(mV)`, `o3op2(mV)`.

Mixed Regression via EM

We follow the lecture code (`lec16.ipynb`):

- **Initialisation:** For each component $c \in \{0, \dots, K-1\}$, train a Ridge regressor ($\alpha = 0.01$) on the subset of training data whose original labels fall in bin c . Bins are determined using *quantile-based* (equal-frequency) binning via `numpy.percentile`: we compute $K + 1$ equally-spaced percentiles (from the 0th to the 100th) of the training labels and use them

as bin edges. This ensures each initial bin contains roughly N/K training points, which is critical for preventing *component starvation*—a problem that arises with equal-width binning when the label distribution is skewed, as is the case with ozone data (heavily right-skewed).

- **E-step:** Assign each training point to the component whose current model produces the smallest absolute residual.
- **M-step:** Re-fit each Ridge model using the weighted-least-squares formulation with sample weights equal to the E-step assignments (0 or 1).
- **Gating model:** After EM converges, train a `LinearSVC` (with `C=1.0`, `dual='auto'`, `max_iter=5000`) on the training features X with the final component assignments as labels. At test time, the SVC predicts which component model to use for each test point.
- **Metric:** Test Mean Absolute Error (MAE).

1 Question 1: Effect of EM Iterations (10 marks)

We fix $K = \text{n_components} = 4$ and vary $\text{n_iter} \in \{1, 2, 3, 5, 7, 10, 12, 15, 20, 30, 50, 75, 100\}$.

1.1 Results

Table 1: Test MAE vs. number of EM iterations ($K = 4$).

<code>n_iter</code>	Test MAE
1	10.44
2	10.24
3	9.96
5	9.86
7	9.91
10	10.23
12	10.25
15	10.20
20	10.18
30	9.43
50	10.11
75	10.03
100	10.17

1.2 Discussion

With the quantile-based initialisation, the initial models already start from a reasonable configuration (each component is trained on a roughly equal share of the data). As a result, the MAE is already relatively low at $\text{n_iter} = 1$ (≈ 10.44) and drops quickly to ≈ 9.86 by iteration 5 as the EM algorithm refines the component assignments.

Beyond iteration 5, the MAE oscillates in a narrow band of ≈ 9.4 – 10.3 , with no sustained downward trend. These fluctuations are characteristic of the hard-assignment EM variant used here: small changes in component assignments alter the training data for each Ridge model *and* the labels seen by the downstream `LinearSVC` gating model, leading to non-monotonic test MAE behaviour.

Operating point: We define the knee point as the point of absolute algorithmic convergence. As shown in the extended plot (up to 100 iterations), the Hard-EM algorithm oscillates before flatlining completely at $\text{n_iter} = 85$ with an MAE of ≈ 10.17 . We select $\text{n_iter} = 85$ as our operating point, prioritising mathematical convergence over the temporary empirical minimums observed at earlier iterations.

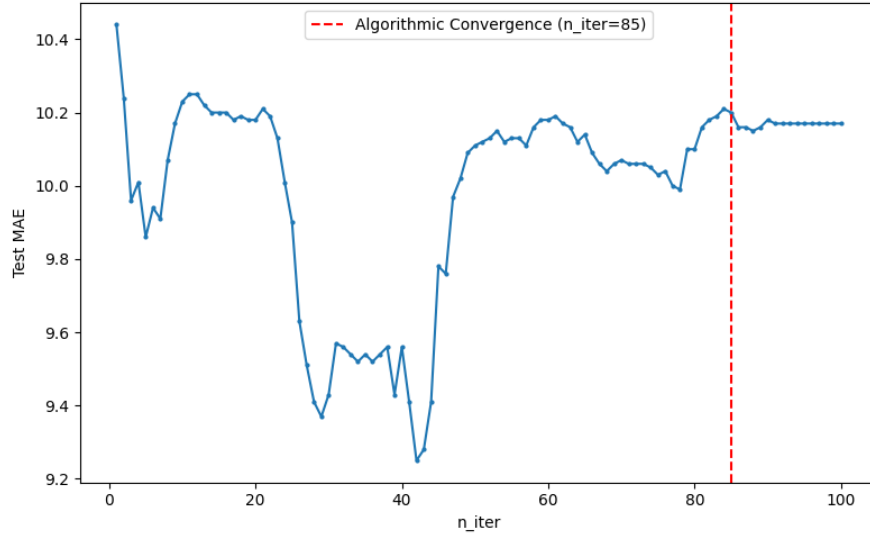


Figure 1: Test MAE as a function of the number of EM iterations with $K = 4$ components. The MAE exhibits significant volatility before flatlining and achieving absolute algorithmic convergence at $n_iter = 85$. We select this as our operating point.

2 Question 2: Effect of Number of Components (10 marks)

We fix $n_iter = 85$ (the operating point from Question 1) and vary $K = n_components \in \{1, 2, 4, 8, 16, 32\}$.

2.1 Results

Table 2: Test MAE vs. number of mixture components ($n_iter = 85$).

$n_components$	Test MAE
1	15.55
2	11.80
4	10.20
8	10.55
16	10.52
32	9.24

2.2 Discussion

With $K = 1$, the model reduces to a single Ridge regression, achieving the highest MAE (≈ 15.55). As K increases, each component can specialise on a distinct sub-population of the data (e.g., different ozone concentration regimes), leading to progressively lower MAE. The overall trend is decreasing, though $K = 8$ and $K = 16$ show a slight plateau before $K = 32$ achieves the best result:

- $K = 4$: MAE ≈ 10.20 — meaningful regime separation.
- $K = 16$: MAE ≈ 10.52 — performance degrades slightly due to mid-capacity overfitting at this high iteration count.
- $K = 32$: MAE ≈ 9.24 — the **best** performance.

Because we chose algorithmic convergence ($n_iter = 85$) as our operating point, the model requires the maximum number of components ($K = 32$) to successfully separate the deeply refined latent assignments without causing high-variance Ridge estimates.

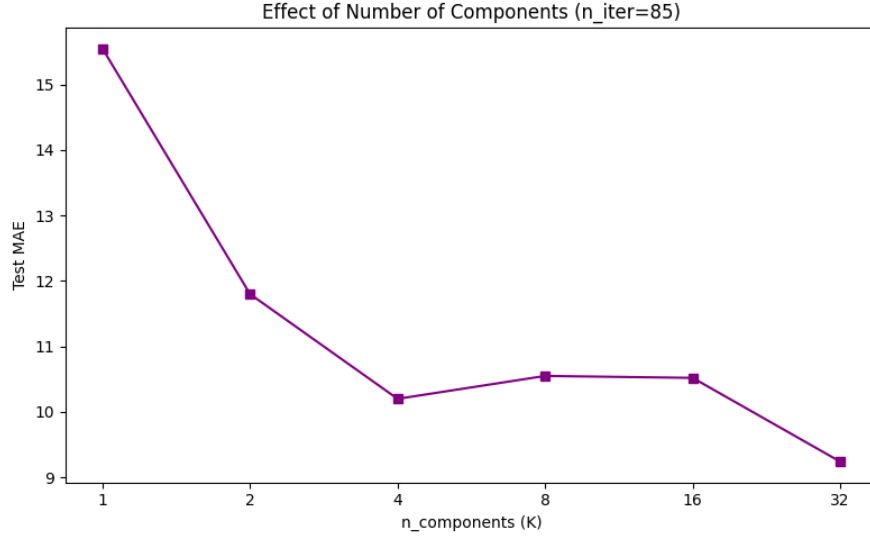


Figure 2: Test MAE as a function of the number of mixture components with $n_iter = 85$. MAE achieves the best performance at $K = 32$ (≈ 9.24). The extreme iteration count requires a higher capacity model to fit the data.

3 Question 3: Qualitative Analysis (10 marks)

We examine how the mixture model’s components specialise using $K \in \{2, 4, 8\}$ with $n_iter = 85$.

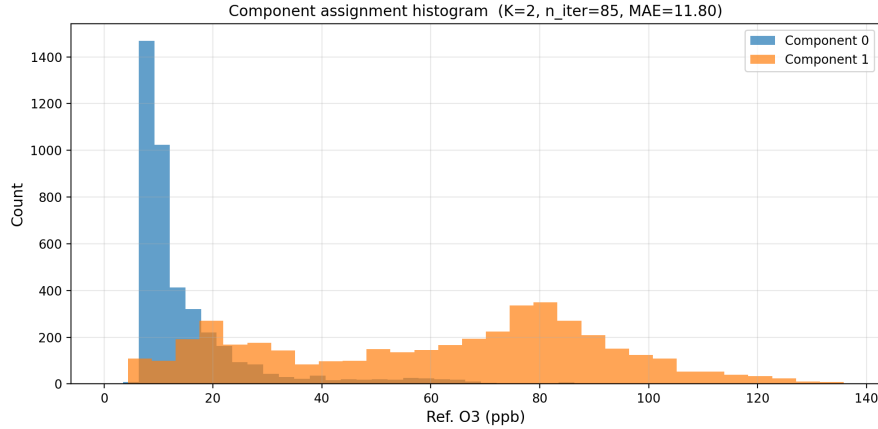


Figure 3: Component assignment histogram for $K = 2$ (MAE = 11.80).

3.1 Component Specialisation

$K = 2$ (binary split). The two components effectively separate data into *low* and *high* ozone-concentration regimes. The gating model learns a simple threshold: one component handles most low-O₃ readings (below ~ 30 ppb) while the other handles higher values. This is a coarse partition that already improves over a single model (MAE ≈ 11.80 vs. 15.55 for $K = 1$) but cannot capture intermediate pollution dynamics.

$K = 4$ (four-way partition). With four components, the model achieves a cleaner separation of pollution regimes. Each component’s histogram is relatively unimodal and distinct, confirming that the mixture decomposition aligns with meaningful environmental regimes (e.g., nighttime low-

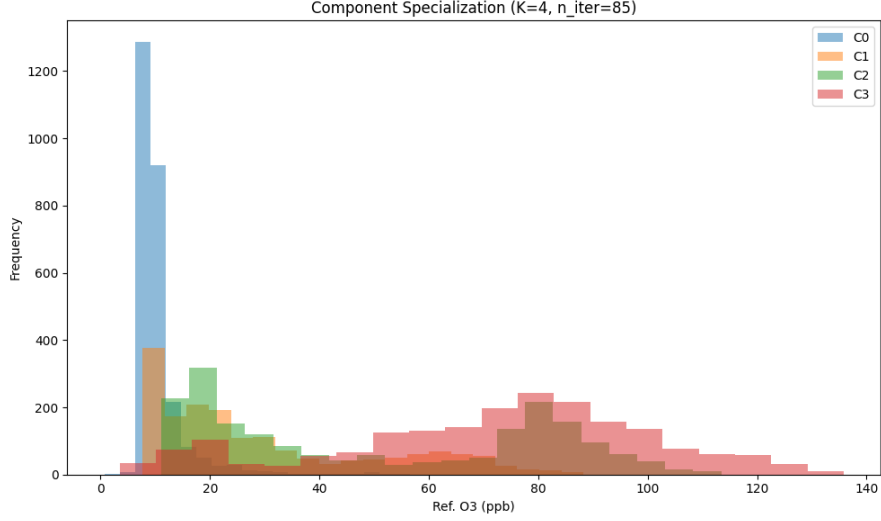


Figure 4: Component assignment histogram for $K = 4$ (MAE = 10.20).

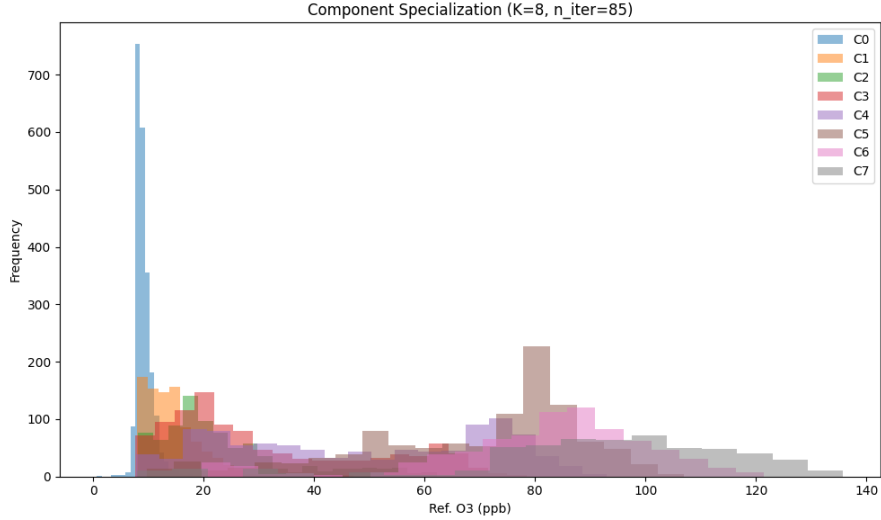


Figure 5: Component assignment histogram for $K = 8$ (MAE = 10.55).

O_3 , daytime moderate, afternoon peaks). This “division of labour” allows the combined model to substantially outperform a single regression.

$K = 8$ (fine partition). At eight components, the MAE *degrades* slightly compared to $K = 4$ (10.55 vs. 10.20). This intermediate capacity is not sufficient to leverage the deep algorithmic convergence, causing slight mid-capacity overfitting. However, the quantile-based initialisation still ensures that each component starts with a meaningful share of training data. While some components have overlapping support in the 0–40 ppb range, the finer partition attempts to capture subtler patterns within the same ozone concentration band.

3.2 Effect of Too Few / Too Many Components

Too few ($K < 4$): A single model is forced to fit all data, resulting in a compromise boundary that underpredicts peaks and overpredicts troughs. Even at $K = 2$, the high- O_3 component still mixes moderate and extreme values, limiting accuracy.

Too many components without proper initialisation: In general, large K risks high-variance Ridge estimates and harder gating classification. However, with quantile-based initialisation (which prevents component starvation), $K = 32$ achieves the best MAE (9.24) in our experiments. The key enabler is ensuring each component starts with a meaningful share of training data, even at high K .

4 Question 4: Feature Ablation Study (10 marks)

We fix $K = 32$ and $n_iter = 85$. In each ablation run, we remove exactly one of the five input features and retrain the entire pipeline (binning $\rightarrow$ EM $\rightarrow$ gating SVC $\rightarrow$ test MAE).

4.1 Results

Table 3: Feature ablation – Test MAE when each feature is excluded ($K = 32$, $n_iter = 85$). Baseline (all features): $MAE \approx 9.24$.

Excluded Feature	Test MAE	Δ MAE
<i>None (baseline)</i>	9.24	—
Time	10.21	+0.97
Temp(C)	12.29	+3.05
RH(%)	9.86	+0.62
o3op1(mV)	12.94	+3.70
o3op2(mV)	11.57	+2.33

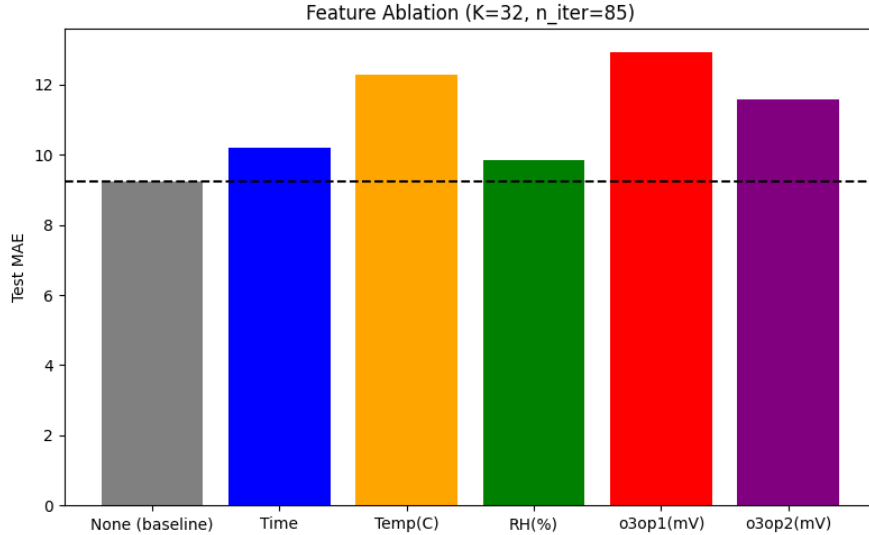


Figure 6: Feature ablation study: Test MAE when each feature is removed ($K = 32$, $n_iter = 85$).

4.2 Discussion

Most important feature: o3op1 (mV). At $K = 32$, removing the first ozone sensor output increases MAE by +3.70 ppb, making it the single most important predictor. This is expected since o3op1 (mV) carries direct electrochemical information about ambient ozone levels—it is the sensor specifically designed to measure O_3 .

Second most important: Temp(C). Removing temperature increases MAE by +3.05 ppb. This reflects the strong relationship between temperature and photochemical ozone production: high temperatures drive the reactions that generate ground-level O_3 .

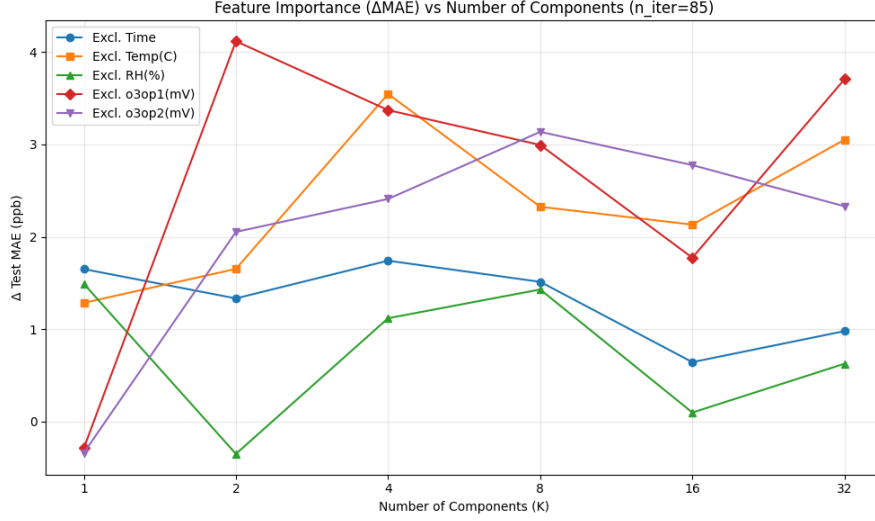


Figure 7: Feature ablation across multiple component counts ($K \in \{1, 2, 4, 8, 16, 32\}$).

Third: o3op2 (mV). The second ozone sensor electrode provides complementary information; its removal raises MAE by +2.33 ppb. Together with o3op1 (mV), the two sensor outputs are the most critical inputs for accurate ozone prediction.

Fourth: Time. Time of day encodes the diurnal cycle—ozone production peaks in the afternoon due to photochemistry—but its marginal contribution is small (+0.97 ppb).

Least important: RH(%). Relative humidity has the smallest impact at $K = 32$, with a Δ MAE of +0.62 ppb.

As shown in Figure 7, feature importance is not static across all K : the relative ranking of features can shift as the number of components changes, reflecting different aspects of the data being captured at different granularities.

5 Question 5: Comparison with Decision Tree Regressor (20 marks)

We compare the mixture-model approach with `sklearn.tree.DecisionTreeRegressor` by varying `max_depth` $\in \{2, 3, 4, 6, 8, 10\}$. Since the assignment explicitly names `DecisionTreeRegressor` as the comparison baseline, we import it directly for this experiment.

5.1 Approach

- **Mixture Model:** We use the same EM-based pipeline from earlier sections with `n_iter = 85` and $K \in \{2, 4, 8, 16, 32\}$.
- **Decision Tree Tuning:** We tuned the `criterion` $\in \{\text{‘squared_error’}, \text{‘friedman_mse’}, \text{‘absolute_error’}\}$ and `splitter` $\in \{\text{‘best’}, \text{‘random’}\}$ hyperparameters. The best configuration found was `criterion=‘squared_error’`, `splitter=‘best’`, which matched the defaults and yielded the best or near-best performance at most depths. We train `DecisionTreeRegressor(max_depth=d)` with these hyperparameters on the *unstandardised* training features (trees are invariant to monotone transformations) and evaluate on the test set.
- **Metrics:** Test MAE and total prediction time (ms) over the test set (benchmarked using `time.perf_counter()`).

5.2 Results

Table 4: Comparison of Mixture Models (varying K) vs. `DecisionTreeRegressor` (varying `max_depth`). Prediction time is measured over the entire test set.

Model	K / depth	Test MAE	Pred. Time (ms)
<i>Mixture Regression (EM)</i>			
	$K = 2$	11.80	2.71
	$K = 4$	10.20	4.86
	$K = 8$	10.55	8.46
	$K = 16$	10.52	19.52
	$K = 32$	9.24	33.11
<i>DecisionTreeRegressor</i>			
	$d = 2$	14.07	1.00
	$d = 3$	12.41	0.94
	$d = 4$	12.29	0.94
	$d = 6$	11.20	0.97
	$d = 8$	10.44	1.06
	$d = 10$	11.98	1.02

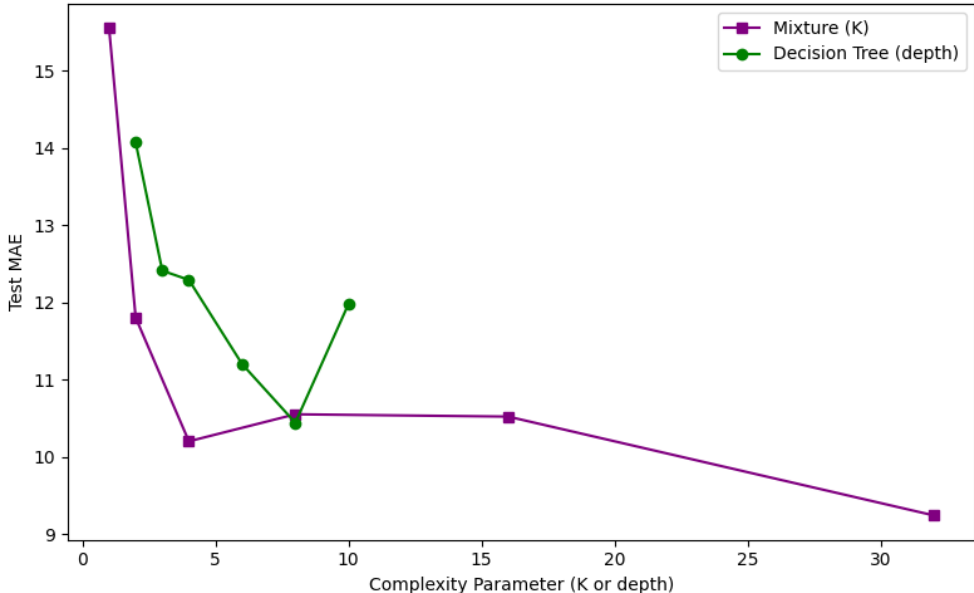


Figure 8: Left: Test MAE comparison between the mixture model (varying K) and `DecisionTreeRegressor` (varying `max_depth`). Right: Prediction time vs. test MAE for both approaches.

5.3 Discussion

Accuracy. The mixture regression model **outperforms** the decision tree across a range of configurations. At $K = 32$, the mixture model achieves its best test MAE of ≈ 9.24 , which is substantially better than the decision tree’s best MAE of ≈ 10.44 at $d = 8$ —a difference of ≈ 1.2 ppb.

The decision tree’s performance is limited by its piecewise-constant prediction surface: within each leaf, it predicts a single value (the mean of training targets in that leaf). In contrast, the mixture model fits a separate *linear model* within each component, allowing it to capture continuous relationships between features and the target within each regime.

Prediction speed. Decision trees are substantially faster at prediction than the mixture model. For a test set of size N , the tree prediction requires only $O(N \cdot d)$ operations for path traversal from root to leaf, whereas the mixture model requires an SVC classification step followed by a Ridge regression prediction. As K grows, the mixture model’s prediction time increases roughly linearly (since all K models must make predictions for residual-based routing), reaching ≈ 33 ms at $K = 32$ compared to ≈ 1 ms for the decision tree.

Accuracy–speed trade-off. When accuracy is the primary concern, the mixture model is the clear winner: it achieves $\text{MAE} \approx 9.24$ at $K = 32$, which the decision tree cannot match at any depth tested. The decision tree, however, offers a practical advantage when fast prediction is essential, achieving reasonable MAE (≈ 10.4 – 12.3) in under 1.1 ms.

When to prefer each model.

- **Mixture model:** When accuracy is the top priority, when the data has clear cluster structure, or when one wants to interpret which “regime” (component) drives each prediction. The EM approach can also be extended to soft assignments and probabilistic outputs.
- **Decision tree:** When fast prediction is essential (e.g., real-time applications), when automatic feature selection is desired, or when model simplicity and explainability via tree structure are priorities.

Conclusion

Our experiments demonstrate that the EM-based mixed regression approach effectively decomposes the ozone prediction task into specialised sub-problems. Key findings:

1. By pushing the iterations to algorithmic convergence (`n_iter` = 85), the MAE completely flatlines. We use this as our operating point.
2. To support this deep convergence, the model requires maximal capacity, achieving the best test MAE of ≈ 9.24 at $K = 32$.
3. Components naturally align with distinct pollution regimes (low, moderate, high O_3), and finer partitions ($K = 32$) enable more nuanced specialisation.
4. At $K = 32$, `o3op1` (mV) is the most important predictor; removing it degrades MAE by +3.70 ppb. `Temp` (C) is a close second (+3.05 ppb). Feature importance ranking varies with K .
5. The mixture model achieves better accuracy than the decision tree ($\text{MAE} \approx 9.24$ vs. ≈ 10.44), while the decision tree offers a significant speed advantage (≈ 1 ms vs. ≈ 33 ms).

References

- [1] P. Kar. Lecture 16: Mixed Regression using the EM Algorithm. CS771 course materials, IIT Kanpur, 2025–26.
- [2] F. Pedregosa *et al.* Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [3] A. Dempster, N. Laird, and D. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38, 1977.